

Factoring Polynomials

Practice Paper — Digital SAT Style

Difficulty Mix: Easy (Q1–12) | Medium (Q13–25) | Hard (Q26–35) · Total: 35 Questions

DIFFICULTY: EASY

Q1. Which is the completely factored form of $x^2 - 9$?

- A) $(x - 3)^2$
 - B) $(x + 3)(x - 3)$
 - C) $(x - 9)(x + 1)$
 - D) $x(x - 9)$
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DIFFICULTY: EASY

Q2. Factor: $x^2 + 7x + 12$

- A) $(x + 3)(x + 4)$
 - B) $(x + 2)(x + 6)$
 - C) $(x + 1)(x + 12)$
 - D) $(x + 4)(x + 3)$
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DIFFICULTY: EASY

Q3. Which is a factor of $x^2 - x - 12$?

- A) $(x - 4)$
 - B) $(x + 4)$
 - C) $(x - 2)$
 - D) $(x + 2)$
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DIFFICULTY: EASY

Q4. Factor out the GCF: $6x^3 + 9x^2 - 3x$

- A) $3(2x^3 + 3x^2 - x)$
 - B) $3x(2x^2 + 3x - 1)$
 - C) $x(6x^2 + 9x - 3)$
 - D) $3x^2(2x + 3)$
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DIFFICULTY: EASY

Q5. Which expression is equivalent to $x^2 - 16$?

- A) $(x - 4)^2$
 - B) $(x + 4)(x - 4)$
 - C) $(x - 2)(x - 8)$
 - D) $(x + 2)(x - 8)$
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DIFFICULTY: EASY

Q6. Factor: $2x^2 + 8x$

- A) $2(x^2 + 4x)$
 - B) $2x(x + 4)$
 - C) $x(2x + 8)$
 - D) $2x(x) + 8x$
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DIFFICULTY: EASY

Q7. What are the solutions of $x^2 - 5x + 6 = 0$?

- A) $x = 2$ and $x = 3$
 - B) $x = -2$ and $x = -3$
 - C) $x = 1$ and $x = 6$
 - D) $x = -1$ and $x = 6$
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DIFFICULTY: EASY

Q8. Which is a perfect square trinomial?

- A) $x^2 + 6x + 9$
 - B) $x^2 + 6x + 6$
 - C) $x^2 - 6x - 9$
 - D) $x^2 + 9$
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DIFFICULTY: EASY

Q9. Factor: $4x^2 - 25$

- A) $(4x - 5)(x + 5)$
- B) $(2x - 5)(2x + 5)$
- C) $(2x - 5)^2$

D) $(4x + 5)(x - 5)$

DIFFICULTY: EASY

Q10. If $x^2 + bx + c = (x + 5)(x - 2)$, what is the value of b ?

- A)** 10
 - B)** 3
 - C)** 7
 - D)** -10
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DIFFICULTY: EASY

Q11. Factor: $x^2 - 10x + 25$

- A)** $(x - 5)^2$
 - B)** $(x + 5)^2$
 - C)** $(x - 5)(x + 5)$
 - D)** $(x - 25)(x + 1)$
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DIFFICULTY: EASY

Q12. Which of the following has $(x + 3)$ as a factor?

- A)** $x^2 - 9$
 - B)** $x^2 + 9$
 - C)** $x^2 + 6x + 9$
 - D)** $x^2 - 6x + 9$
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DIFFICULTY: MEDIUM

Q13. Factor completely: $2x^2 + 5x - 3$

- A)** $(2x - 1)(x + 3)$
 - B)** $(2x + 1)(x - 3)$
 - C)** $(2x - 3)(x + 1)$
 - D)** $(x + 3)(2x - 1)$
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DIFFICULTY: MEDIUM

Q14. Factor: $x^3 - 4x$

- A)** $x(x^2 - 4)$

- B) $x(x - 2)^2$
 - C) $x(x - 2)(x + 2)$
 - D) $(x^2 - 2)(x + 2)$
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DIFFICULTY: MEDIUM

Q15. What are the solutions of $3x^2 - 12x = 0$?

- A) $x = 0$ only
 - B) $x = 4$ only
 - C) $x = 0$ and $x = 4$
 - D) $x = 0$ and $x = -4$
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DIFFICULTY: MEDIUM

Q16. Factor: $6x^2 - 7x - 3$

- A) $(2x - 3)(3x + 1)$
 - B) $(3x + 1)(2x - 3)$
 - C) $(6x + 1)(x - 3)$
 - D) $(2x + 1)(3x - 3)$
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DIFFICULTY: MEDIUM

Q17. Solve for x : $x^2 + 3x - 28 = 0$

- A) $x = 4$ and $x = -7$
 - B) $x = -4$ and $x = 7$
 - C) $x = 4$ and $x = 7$
 - D) $x = -4$ and $x = -7$
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DIFFICULTY: MEDIUM

Q18. Which factored form represents $4x^2 - 12x + 9$?

- A) $(4x - 3)(x - 3)$
 - B) $(2x - 3)^2$
 - C) $(2x - 9)(2x - 1)$
 - D) $(4x - 9)(x - 1)$
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DIFFICULTY: MEDIUM

Q19. Factor: $x^3 + 2x^2 - x - 2$

- A) $(x + 2)(x - 1)(x + 1)$
 - B) $(x - 2)(x + 1)^2$
 - C) $(x + 2)(x^2 - 1)$
 - D) $(x - 2)(x^2 + 1)$
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DIFFICULTY: MEDIUM

Q20. If $x^2 - kx + 16 = (x - 4)^2$, what is the value of k ?

- A) 4
 - B) 8
 - C) 16
 - D) -8
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DIFFICULTY: MEDIUM

Q21. What are the solutions of $2x^2 - 3x - 2 = 0$?

- A) $x = -1/2$ and $x = 2$
 - B) $x = 1/2$ and $x = -2$
 - C) $x = 2$ and $x = -2$
 - D) $x = 1/2$ and $x = 2$
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DIFFICULTY: MEDIUM

Q22. Factor: $9x^2 - 6x + 1$

- A) $(3x + 1)^2$
 - B) $(3x - 1)^2$
 - C) $(9x - 1)(x - 1)$
 - D) $(3x - 1)(3x + 1)$
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DIFFICULTY: MEDIUM

Q23. Solve: $x(x - 5) = 14$

- A) $x = -2$ and $x = 7$
 - B) $x = 2$ and $x = -7$
 - C) $x = 5$ and $x = 14$
 - D) $x = 7$ and $x = 2$
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DIFFICULTY: MEDIUM

Q24. Factor: $12x^2 + 8x - 15$

- A) $(6x - 5)(2x + 3)$
 - B) $(4x - 3)(3x + 5)$
 - C) $(6x + 5)(2x - 3)$
 - D) $(3x - 5)(4x + 3)$
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DIFFICULTY: MEDIUM

Q25. The expression $x^3 - 81$ factors as:

- A) $(x^2 - 9)^2$
 - B) $(x - 3)^3$
 - C) $(x^2 - 9)(x^2 + 9)$
 - D) $(x - 3)(x + 3)(x^2 + 9)$
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DIFFICULTY: HARD

Q26. For what value of c is $x^2 + 12x + c$ a perfect square trinomial?

- A) 6
 - B) 24
 - C) 36
 - D) 144
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DIFFICULTY: HARD

Q27. Factor completely: $2x^3 - 16$

- A) $2(x - 2)(x^2 + 2x + 4)$
 - B) $2(x^3 - 8)$
 - C) $(2x - 4)(x^2 + 2x + 4)$
 - D) $2(x - 2)^3$
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DIFFICULTY: HARD

Q28. Solve: $x^2 - 4x = 21$

- A) $x = 7$ and $x = 3$
- B) $x = 7$ and $x = -3$
- C) $x = -7$ and $x = 3$

D) $x = -7$ and $x = -3$

DIFFICULTY: HARD

Q29. Which of the following is a factor of $x^3 + x^2 - 4x - 4$?

A) $(x - 1)$

B) $(x + 1)$

C) $(x - 2)$

D) $(x + 4)$

DIFFICULTY: HARD

Q30. The solutions of $ax^2 + bx + c = 0$ are $x = 3/2$ and $x = -4$.

If $a = 2$, which equation is correct?

A) $2x^2 + 5x - 12 = 0$

B) $2x^2 - 5x + 12 = 0$

C) $2x^2 + 5x + 12 = 0$

D) $2x^2 - 5x - 12 = 0$

DIFFICULTY: HARD

Q31. Factor: $x^4 - 5x^2 + 4$

A) $(x^2 - 1)(x^2 - 4)$

B) $(x - 1)(x + 1)(x - 2)(x + 2)$

C) $(x^2 - 4)^2$

D) Both A and B

DIFFICULTY: HARD

Q32. If $x = -3$ is a double root of $x^3 + ax^2 + bx + c = 0$ and $x = 1$ is the third root,

what is the value of a ?

A) 5

B) 7

C) 9

D) 11

DIFFICULTY: HARD

Q33. For which values of k does $kx^2 - 8x + 4 = 0$ have exactly two distinct real solutions?

- A) $k < 4$
 - B) $k > 4$
 - C) $k < 4$ and $k \neq 0$
 - D) $k > 0$
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DIFFICULTY: HARD

Q34. Factor: $8x^3 + 27$

- A) $(2x + 3)(4x^2 - 6x + 9)$
 - B) $(2x + 3)(4x^2 + 6x + 9)$
 - C) $(2x - 3)(4x^2 + 6x + 9)$
 - D) $(2x + 3)^3$
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DIFFICULTY: HARD

Q35. The polynomial $x^3 - 7x + 6$ has $x = 1$ as a root. What are the other two roots?

- A) $x = 2$ and $x = -3$
 - B) $x = -2$ and $x = 3$
 - C) $x = 3$ and $x = -3$
 - D) $x = 2$ and $x = 3$
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